



Augmenting urban planning with computer vision: A review of the state-of-the-art

Raveena Marasinghe^a, Tan Yigitcanlar^{a,*}, Xinyu Fu^b, Steven Jige Quan^c

^a QUT Urban AI Hub, School of Architecture and Built Environment, Queensland University of Technology, Brisbane, QLD, Australia

^b Department of Landscape Architecture and Urban Planning, Texas A&M University, College Station, TX, USA

^c Department of Environmental Planning, Seoul National University, Seoul, Republic of Korea

ARTICLE INFO

Keywords:

Computer vision
Artificial intelligence
Convolutional neural networks
Foundation models
Built environment analysis
Urban planning

ABSTRACT

Computer Vision (CV) offers powerful methods for analysing built environments, yet its integration into planning workflows remains fragmented. This study conducts a systematic review of CV applications in urban planning using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol, synthesising prevailing models, methods, and application domains. The review identifies five dominant CV research clusters in urban contexts (built environment analysis, urban sensing and data acquisition, smart mobility, methodological advances, and 3D urban modelling) and synthesises that CV is primarily used to extract physical and perceptual urban features, support diagnostic analytics, and enable emerging visualisation and generative design workflows. Advances in deep learning, convolutional neural networks, and emerging foundation models (including vision transformers and diffusion-based generative systems) are expanding capability across diverse CV tasks and lowering barriers to adoption. Building on an inductive synthesis of application patterns across the reviewed studies, the paper proposes an Urban Visual Intelligence (UVI) hierarchy that clarifies how imagery is translated into progressively more planning-relevant outputs, from observation and diagnosis to associative explanation, forecasting, and intervention prototyping. The review also highlights constraints related to data representativeness, transferability, interpretability, privacy and generative unreliability, underscoring the need for stronger validation and governance as outputs become more decision relevant. Overall, the review consolidates how CV is reshaping urban analytics and offers a structured basis for responsible integration into planning practice.

1. Introduction and background

The integration of artificial intelligence (AI) has increasingly influenced urban planning and design, with computer vision (CV) expanding analytical capabilities beyond traditional methods (Faisal et al., 2021; Wael et al., 2022; Marasinghe et al., 2024). CV enables computational interpretation of images and video by extracting meaningful information from visual data and has become central to modern image analysis across disciplines (Wiley & Lucas, 2018; Matsuzaka & Yashiro, 2023). In urban contexts, advances in neural networks and deep learning (DL) have improved CV performance in built-environment assessment, supporting scalable interpretation of streetscapes, infrastructure, land cover, and mobility environments (Sanchez, 2023; Zhao et al., 2023; Yigitcanlar et al., 2024).

CV and machine learning (ML), have enabled the systematic

extraction of large, time-intensive urban datasets at greater spatial scale, revealing previously hard-to-measure city attributes and supporting computational perspectives on complex urban dynamics (Ibrahim et al., 2021; Payntar, 2022; Suel et al., 2023). Studies increasingly use street view images (SVIs), to support planning-relevant tasks, including built-environment measurement (Knura et al., 2021), automated detection and monitoring of urban change, prediction of street-network configurations (Fang et al., 2022), estimation of perceptual qualities (Song et al., 2022a), and analysis of socioeconomic dynamics such as urbanisation patterns and population activity proxies (Koch et al., 2021).

As cities face intensifying pressures related to sustainability, mobility, public health, climate exposure, and inequality, the ability to generate fine-grained, and timely evidence from visual data is increasingly relevant to urban planning (Metaxiotis et al., 2010). Responding to

* Corresponding author.

E-mail address: tan.yigitcanlar@qut.edu.au (T. Yigitcanlar).

<https://doi.org/10.1016/j.scs.2026.107209>

Received 24 March 2025; Received in revised form 3 February 2026; Accepted 3 February 2026

Available online 4 February 2026

2210-6707/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

the growing integration of CV in urban contexts, this paper provides a comprehensive review of CV applications in urban planning, synthesising macro-level research clusters with micro-level application patterns and planning-oriented analytical insights. While prior reviews have provided valuable perspectives on CV methods and application domains (such as knowledge-based techniques (Matsuzaka & Yashiro, 2023), planning-focused applications (Marasinghe et al., 2024), traffic environment perception (Azfar et al., 2024), and built environment sensing (Wang et al., 2024)) fewer studies explicitly organise CV research around planning functions, levels of decision relevance and technical ecosystems. Beyond a descriptive inventory of methods, the review connects technical capabilities to how visual evidence is produced, interpreted, and operationalised within planning practice.

Accordingly, this review provides: (a) a broader CV-urban research landscape by mapping dominant clusters and (b) provides an in-depth synthesis of planning-relevant workflows. The paper (c) consolidates the commonly used CV tasks, models, and training datasets that underpin these applications. Building on an inductive synthesis of recurring application patterns, the review proposes (d) an Urban Visual

Intelligence (UVI) hierarchy that shows how CV can systematically transform raw imagery into multi-level spatial intelligence (from measurement and diagnosis to associative explanation, forecasting, and design intervention support) offering a structured approach through which visual evidence is translated into urban planning insights. This integrated perspective supports both researchers and planners by clarifying analytical choices, planning relevance, and constraints affecting responsible adoption.

2. Research design

This study adopts a systematic literature review methodology guided by the PRISMA protocol. The review aims to deepen understanding of how CV contributes to urban planning by synthesising current applications and methodological approaches that inform planning practice. The guiding research questions are: (a) How is CV currently applied in urban planning context? (b) What models and datasets are the most commonly used in CV applications within urban contexts?

A PRISMA-based review was conducted in four stages: Identification,

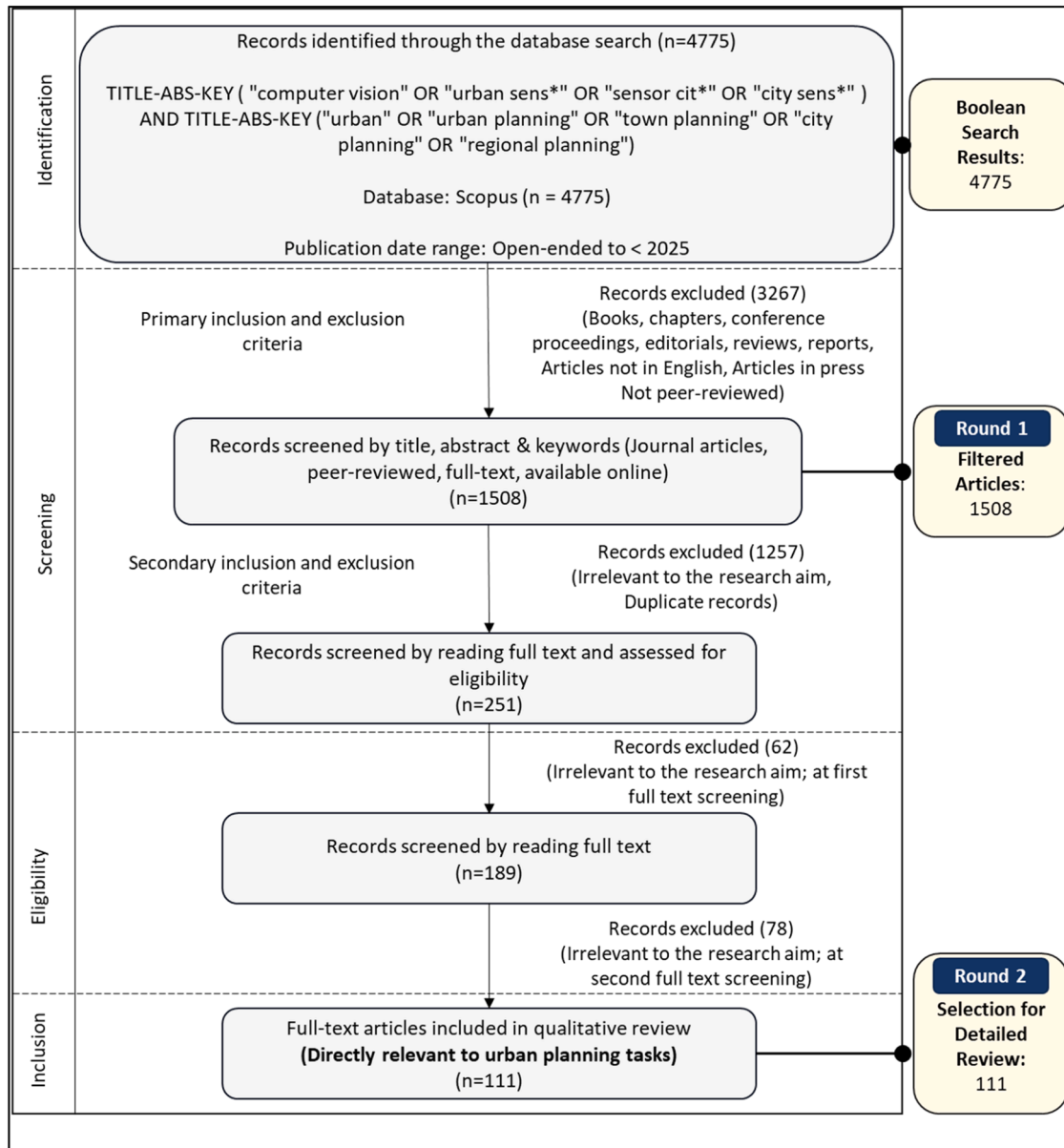


Fig. 1. The process of the systematic PRISMA review.

Screening, Eligibility, and Inclusion (Fig. 1). During the Identification stage, relevant literature was retrieved from the Scopus database, which provides broad coverage of peer-reviewed publications across disciplines and is widely regarded as reliable for systematic reviews, although future research could incorporate additional databases. A deliberately constructed Boolean function was used to retrieve articles covering a variety of CV applications in the context of urban planning. The utilised Boolean operation is as follows, “(TITLE-ABS-KEY (“computer vision” OR “urban sens*” OR “sensor cit*” OR “city sens*”) AND TITLE-ABS-KEY (“urban” OR “urban planning” OR “town planning” OR “city planning” OR “regional planning”))”. The search was conducted up to the end of 2024 and yielded 4775 records for screening. The scope and rationale for these search terms is detailed below.

Defining the scope of “urban planning” is essential for identifying relevant papers. While we employed keywords such as ‘urban’, urban planning’ to capture a broad range of studies, our analysis was not limited to a narrow definition of urban planning. In this review, “urban planning” extends beyond traditional city planning to encompass the broader processes, including articles related to urban analytics, spatial modelling, design, and decision-making tasks that contribute to or are directly associated with urban planning. The terms “urban planning” and “computer vision” are broad by design to ensure the inclusion of a diverse range of studies, while terms like “urban sens*” and “city sens*” though broader than the specific focus on CV, highlight the growing role of sensor-based technologies in monitoring and understanding urban environments, enriching the pool of applications for analysis.

The review acknowledges that many urban studies employing CV technology also use terms such as “image processing,” “artificial intelligence,” “machine learning,” “deep learning,” and “neural networks.” However, in this study, “computer vision” was selected as the central term to maintain focus on its specific application in visual data analytics. This approach maintains clarity and consistency in the scope of the review. Despite this focused approach, a significant amount of pertinent literature was identified by conducting the search across all title, abstract, and keyword fields, fulfilling the study’s objectives. This broader approach allowed us to develop a comprehensive understanding of the role that CV plays across various facets of urban research and practice.

The Screening stage involved the application of inclusion and exclusion criteria to successfully filter the journal papers (Table 1). The study included only peer-reviewed journal articles that were published in English, and in alignment with the research goal. This led to a reduction in the number of records to 1508. The 1508 filtered articles (Fig. 1) were analysed using VOSviewer software to gain a general understanding of the distribution of research clusters within the literature.

After reviewing these records’ titles, abstracts, and keywords for validity, 251 articles were selected for the Eligibility stage (Fig. 1). During the Eligibility and Inclusion stages, each paper was assessed for alignment with the review objective. Following a second round of full-text screening, 111 pertinent papers were selected for in-depth qualitative synthesis.

Table 1
Criteria for inclusion articles.

Criterion for inclusion	Criterion for exclusion
Published in journals with peer review	Books, book sections, conference papers, editorial articles, review papers, and reports
Written in English	Articles not in English
Directly related to the research questions	Not directly associated with the research questions
Published prior to 2025	Articles in press

3. Analysis and results

3.1. General observations

This section reports the results of the review in two layers. First, the broader set of screened records ($n = 1508$) is analysed to map dominant research clusters and thematic structure. Second, the final set of planning-relevant studies ($n = 111$) is synthesised to examine how CV is applied in urban planning tasks, which CV tasks and models are used, and which training datasets most frequently underpin these applications.

3.1.1. Classification of literature records by research clusters

The screening process reflects the rapid growth of CV research in urban contexts, particularly after 2020 associated with advances in DL and increased availability of large-scale visual datasets (Wu et al., 2025; Matsuzaka & Yashiro, 2023).

The 1508 filtered articles were analysed using VOSviewer to examine the distribution of research clusters within the literature. Keywords from each record, as provided by the Scopus database, were entered into VOSviewer, and those with at least five occurrences were selected for visualisation. In the resulting Cluster Network Map (Fig. 2), each node represents a keyword, with node size reflecting keyword frequency, and links indicating co-occurrence relationships between keywords. Clustering of keywords was automated by VOSviewer and subsequently labelled by the authors based on dominant themes.

Five distinct clusters were identified (Table 2): (a) Urban Planning and Built Environment, (b) Urban Sensing and Data Acquisition, (c) Smart Mobility and Transportation, (d) Methods and Techniques, and (e) 3D Modelling and Urban Reconstruction. These keyword clusters capture the thematic structure of CV research as applied to urban contexts and reflect both application-oriented and method-driven strands of the literature.

Clusters 1 and 2 primarily encompass keywords central to urban planning, sustainable development, human activity, land use, environmental monitoring, and spatial analysis (Table 2). Cluster 1 reflects the application of CV in analysing urban form, land use patterns, environmental impacts, and public spaces, emphasising the contribution of visual data to planning and policy-relevant evidence generation. Cluster 2 highlights the integration of sensing technologies, Internet of Things (IoT) systems, and participatory sensing for real-time data collection and decision support in smart city contexts. Together, these clusters indicate how CV has become embedded in core planning functions related to urban diagnostics and spatial assessment enabling data-driven approaches for sustainable urban development.

Cluster 3 focuses on mobility and transport-related applications, including traffic monitoring, pedestrian safety, intelligent transportation systems, and automated surveillance; illustrating the growing role of visual analytics in managing urban movement systems. Cluster 4 comprises keywords on DL architectures, segmentation algorithms, remote sensing and convolutional neural networks (CNNs) that enable advanced image classification, feature extraction, and object detection—underpinning the analytical backbone of CV in urban contexts.

Cluster 5 emphasises 3D vision techniques such as image processing, three-dimensional computer graphics, photogrammetry, LiDAR, and immersive visualisation that support spatial reconstruction, urban modelling, and design-oriented representation of cities. Together, these clusters reveal the multidisciplinary nature of CV in urban planning, spanning from foundational technical methods to practical, application-specific solutions for more responsive, inclusive, and data-informed urban development.

The overlay visualisation (Fig. 3) illustrates a discernible temporal progression in research focus. Earlier studies utilise CV for natural environmental monitoring (Yigitcanlar, 2009), where more recent studies increasingly concentrate on built environment analysis, spatial diagnostics, urban sensing, and advanced CV methodologies, indicating

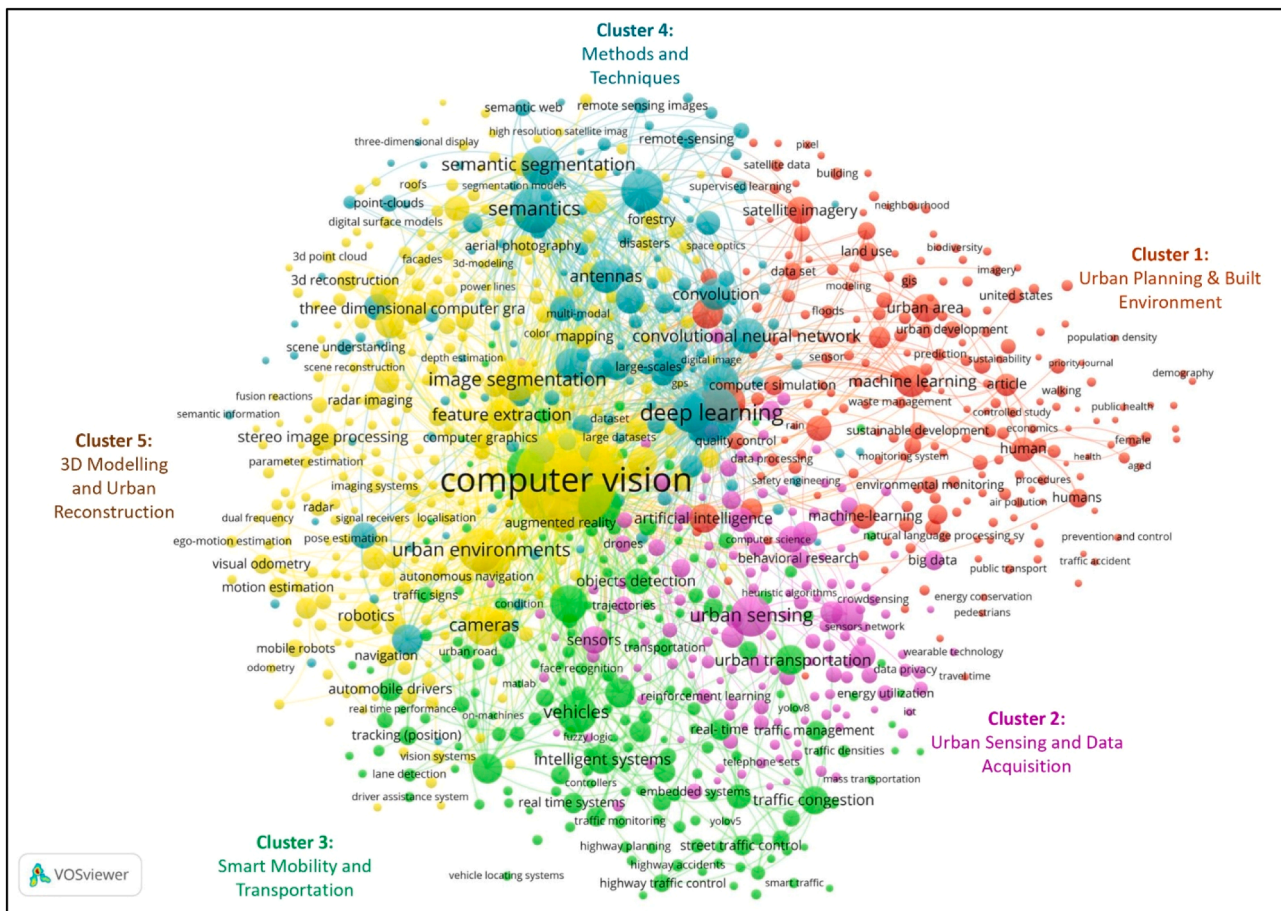


Fig. 2. Cluster network map of literature record (based on keyword occurrences).

a shift towards leveraging CV not only for basic object detection but also for spatial decision-making and scenario analysis, supporting a more holistic understanding of urban systems. The emergence of cutting-edge models and multimodal data sources suggests a growing emphasis on actionable and real-time insights that can inform sustainable urban planning practices across multiple domains.

The density visualisation (Fig. 4) further confirms the concentration of research around DL, semantic segmentation, urban environments, object detection, CNNs, smart city and urban sensing themes, reflecting the convergence of technical development and urban applications. The prominence of keywords such as satellite imagery, remote sensing, UAVs, and aerial photography (largely in Clusters 2 and 5) underscores the growing integration of earth observation and remote sensing data into CV-driven urban analytics. Collectively, the visualisations affirm the methodological evolution and interdisciplinary convergence characterising the current state of CV-based urban research.

Building on these results and informed by critical screening, 111 articles were selected for detailed review and synthesis. In this review, urban planning tasks are understood broadly to include not only formal plan-making but also activities integral to planning processes, such as issue identification, data collection, spatial analysis, design visualisation, and participatory engagement. The following sections synthesise how CV is applied across these tasks, which CV techniques are used, and how outputs are integrated into planning-relevant workflows.

3.2. CV applications in urban planning

The word cloud analysis of keywords from the 111 reviewed articles offers an overview of dominant themes in CV applications in urban planning. Prominent terms such as urbanisation, streetscapes,

environment, mapping, sensing, planning, visualisation and design highlights the core topics of interest in applying CV to urban contexts. Keywords related to perception, modelling, data, surveys, computation and detection further reflect the use of CV to extract planning-relevant information from visual data. The presence of technical terms, such as classification, regression, neural network, feature extraction, and mapping, reflects the methodological foundations underpinning these applications. Collectively, the word cloud illustrates the interdisciplinary nature of CV in urban planning, spanning analytical, diagnostic, and design-oriented functions. These application domains are examined in greater detail in the subsections below. [Fig. 5](#)

3.2.1. Visual and perceptual feature analysis and prediction

A substantial body of research applies CV to analyse built-environment characteristics and streetscape composition (Fig. 6). Studies use CV to extract spatial attributes such as urban character (Koch et al., 2021), traffic infrastructure (Knura et al., 2021), land use and urban changes (Guzder-Williams et al., 2023; Stalder et al., 2024), walkability assessments (Doiron et al., 2022), perceptual attributes (Biljecki, 2023) and activity patterns (Zhang et al., 2022) to support various spatial assessments. Besides, CV has been instrumental in testing correlations between physical spatial features and perceptual attributes, spatial preference, and human experience (Biljecki, 2023).

Furthermore, studies focus on assessing real estate values (Xu et al., 2022), predicting urban growth (Jaad & Abdelghany, 2021), and exploring relationships between spatial characteristics and various socioeconomic factors such as crime (Su et al., 2023) and health (Zhang et al., 2023) have employed CV technology to explore such urban issues. By enabling the prediction of both physical and perceptual qualities of urban space, CV supports context-sensitive and evidence-based

Table 2
Clusters of literature records.

Cluster no	Category name	Colour code	Top Occurring Keywords Examples	No of keywords
1	Urban Planning and Built Environment	Red	urban planning, machine learning image analysis, urban area satellite imagery, urban growth, human, air quality, land use, sustainable urban development, environmental monitoring, urban population, built environment, GIS, pedestrian, urbanization, city, spatial analysis, floods, climate change, public space, prediction, urban transport, building spatiotemporal analysis, perception, neighbourhood, pollution	213
2	Urban Sensing and Data Acquisition	Purple	urban sensing, smart city, sensors decision making, internet of things, forecasting, data acquisition, information management, big data, behavioural research, data mining, trajectories, drones, data visualisation, urban computing, crowdsourcing, smartphones, cost effectiveness, surveys, participatory sensing, real time, data privacy	155
3	Smart Mobility and Transportation	Green	object detection, vehicles, roads and streets, urban transportation, intelligent transportation systems, security systems, traffic control, pedestrian safety, real-time, video cameras, machine-vision, video surveillance, face recognition, license plates	184
4	Methods and Techniques	Blue	deep learning, semantic segmentation, remote sensing, convolutional neural networks, antennas, deep neural networks, image classification, autonomous driving, unmanned aerial vehicles (UAV), large dataset, network architecture, learning algorithms, features extraction, pixels, statistical tests, complex networks, aerial images, point-clouds	140
5	3D Modelling and Urban Reconstruction	Yellow	computer vision, urban environments, image segmentation, cameras, image processing, feature extraction, algorithms, three-dimensional computer graphics,	308

Table 2 (continued)

Cluster no	Category name	Colour code	Top Occurring Keywords Examples	No of keywords
			optical radar, robotics, 3d reconstruction, global positioning system, photogrammetry, virtual reality, mapping, visualization, stereo vision, augmented reality, edge detection, textures, 3d modelling	

planning. In sum, CV's predictive and descriptive capabilities have significantly expanded the spatial intelligence available to urban planners, supporting more targeted and responsive interventions.

3.2.2. Human detection and prediction analysis

CV-based detection and tracking techniques are widely used to analyse human activity in urban spaces, particularly in studies of traffic flow, pedestrian movement, and public space usage. Applications include human action detection, facial recognition, and trajectory analysis (Wong et al., 2021), often combined with clustering techniques, to study activity patterns (Salazar-Miranda et al., 2023), pedestrian trajectories, urban density and vitality (Niu et al., 2022), urban design efforts, user experiences and satisfaction (Wael et al., 2022). These methods allow for a deeper understanding of urban dynamics and help inform better urban design and planning practices. By offering real-time, fine-grained data on human movement and behaviour, CV plays an instrumental role in designing people-centric and adaptive urban environments.

3.2.3. Design visualisation and automated design

Beyond analysis and monitoring, CV is increasingly used for design visualisation and automated design support. Applications include immersive scene construction (e.g., VR/AR) (Luo et al., 2022a), 3D urban modelling and reconstruction (Verma et al., 2021; Liu et al., 2024), as well as supporting real-time feedback and interactive design efforts (Yuan et al., 2023). Additionally, CV has been integrated into generative designs workflows, including automated urban scene generation (Joglekar et al., 2020), text-to-image realistic urban imagery generation (Wang et al., 2025; Seneviratne et al., 2022) and street network design (Fang et al., 2022), further expanding its role in urban planning. These applications mark a shift from passive analysis to active design facilitation, positioning CV as a tool that augments creative exploration and supports iterative planning processes.

3.2.4. Urban visual intelligence through computer vision

Zhang et al. (2024) highlights the promise of 'visual intelligence' for observing and interpreting urban environments employing AI-driven tools. Building on this general idea, the present study develops an UVI hierarchy through CV (Fig. 7) derived inductively from the application patterns identified across the 111 reviewed articles (Sections 3.2.1–3.2.3; Fig. 6). The framework is derived from recurring ways in which reviewed studies convert urban imagery into planning-relevant outputs: (a) measuring and inventorying urban elements, (b) diagnosing patterns and dynamics, (c) linking urban form to human, environmental, or socioeconomic outcomes, (d) forecasting scenarios and impacts, and (e) generating and testing design interventions.

UVI is conceptualised as a hierarchical progression of intelligence, where each level increases the decision-relevance of visual data and addresses a distinct limitation ('intelligence gap') in traditional urban planning practice. The levels are hierarchical in the sense that higher levels typically depend on outputs from lower levels and provide increasing decision-relevance for planning practice. Importantly, each level addresses a specific intelligence gap in conventional planning

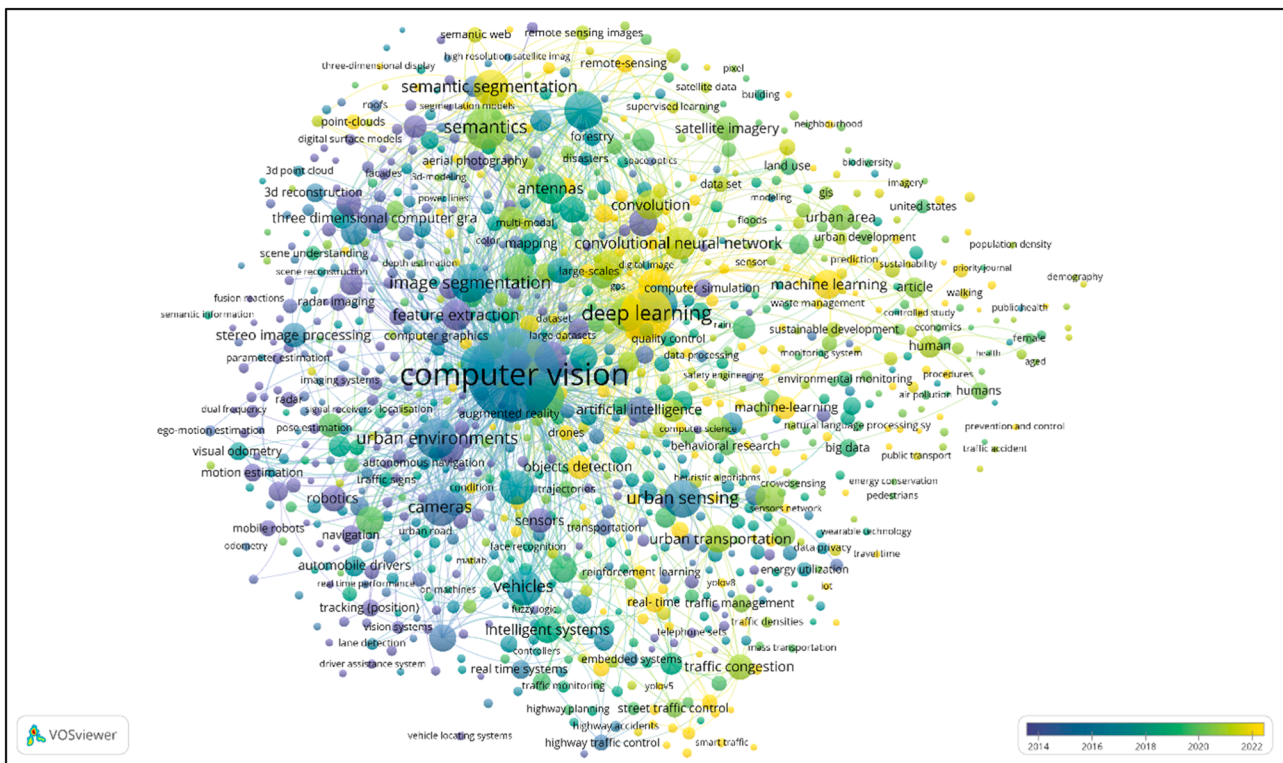


Fig. 3. Overlay visualisation of average publication year on research cluster network map.

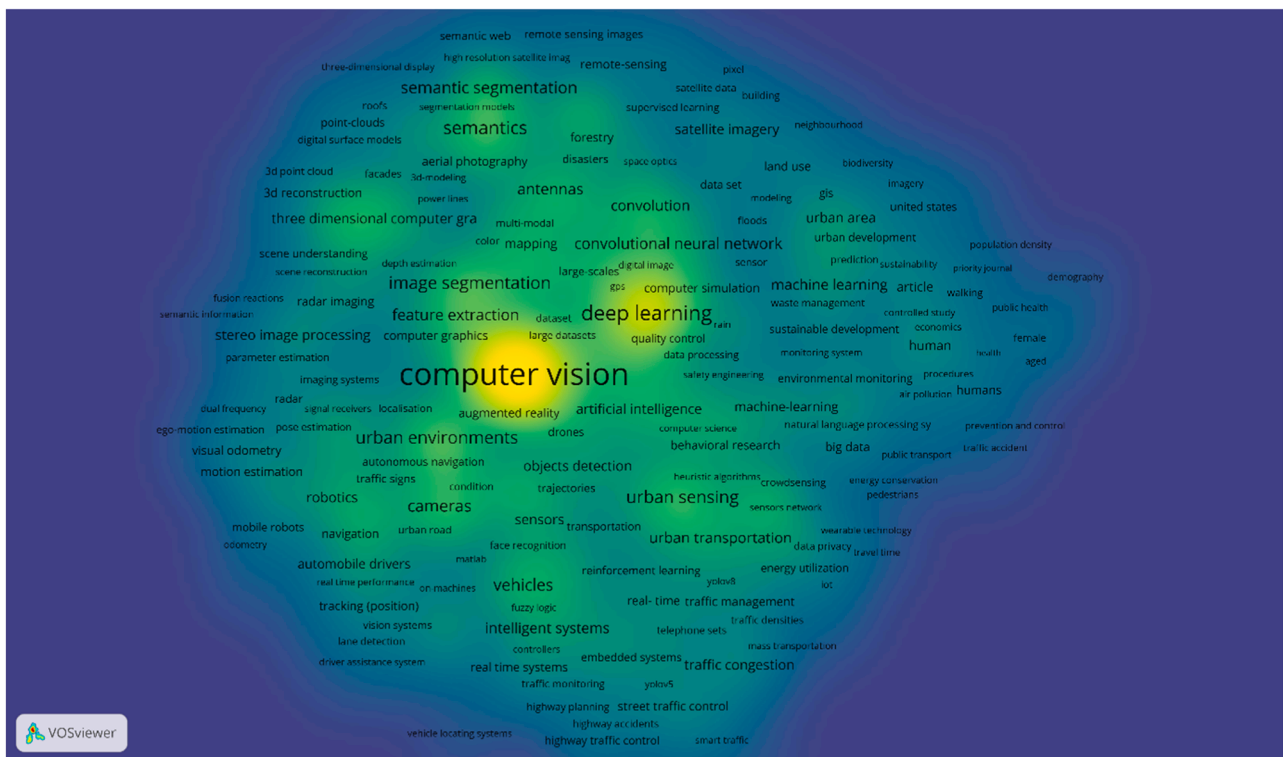


Fig. 4. Keyword occurrence density visualisation map.

workflows.

This culminates in leveraging multi-scale spatial intelligence for evidence-based decision support in planning and policy. This conceptualisation addresses a key challenge in urban visual analysis, where

the vast potential of imagery is often limited by the absence of systematic methods for interpretation and the insufficient integration of extracted data into classical urban theory.

Observational Intelligence: Traditional planning often relies on manual



Fig. 5. Word cloud analysis of keywords in reviewed papers.

audits, limited surveys, infrequent censuses, and coarse land-use inventories. Observational intelligence addresses this gap by using CV to capture and quantify urban features at scale, including streetscape elements, building typologies, greenery, infrastructure, and basic human/vehicle presence using multi-modal imagery (e.g., street-level, aerial, remote sensing). The output is a measurable, spatially referenced evidence base that can update baseline conditions more frequently than conventional approaches.

Analytical Intelligence: Conventional planning diagnosis frequently depends on aggregate indicators and static maps that obscure micro-scale variation and temporal dynamics. Analytical intelligence addresses this gap by enabling fine-grained classification and pattern detection (e.g., segmentation, detection, tracking), supporting diagnostic tasks such as identifying spatial hotspots, monitoring activity flows, and detecting environmental or infrastructural conditions across time. This level shifts visual data from “inventory” to diagnosis.

Correlative Intelligence: A recurring limitation in planning is the difficulty of empirically testing relationships between built form and social/environmental outcomes at scale. Correlative intelligence addresses this gap by linking CV-derived features to perceptual, behavioural, socioeconomic, and environmental variables (e.g., perceived safety, health, crime, housing value, thermal exposure). This level transforms visual measurement into interpretable evidence about associations and potential mechanisms, enabling planners to evaluate how urban form relates to lived experience and sustainability outcomes.

Predictive Intelligence: Traditional scenario analysis and forecasting are often constrained by limited input data, modelling assumptions, and the cost of updating baseline evidence. Predictive intelligence addresses this gap by using CV-derived variables as inputs to forecasting and simulation (e.g., growth prediction, impact estimation, behaviour prediction, risk modelling). This level enables more timely scenario testing and provides forward-looking evidence to support policy appraisal and prioritisation.

Generative Intelligence: Conventional planning and urban design often face a bottleneck in rapidly exploring alternatives, visualising impacts,

and iterating with stakeholders. Generative intelligence addresses this gap by using generative CV (e.g., generative adversarial networks (GANs), diffusion models) and interactive visualisation (VR/AR, HCI systems) to produce design prototypes, visual alternatives, and adaptive interventions. This stage shifts CV from analysis to co-creative planning support, expanding the solution space and enabling participatory engagement, while also requiring stronger safeguards against bias, hallucination, and value misalignment.

In summary, Fig. 7 synthesises the reviewed literature into a staged framework that clarifies how CV can evolve from measurement tools to decision-support and intervention design systems. It provides a systematic and holistic conceptualisation for understanding the physical environment at various scales through CV. The hierarchy also clarifies why higher levels demand stronger validation, transparency, and governance, because the consequences of error increase as outputs move closer to policy and design action Table 3.

3.3. CV tasks, models and algorithms in literature

CV has been applied across a diverse set of tasks in urban studies, each supporting distinct analytical and planning functions. This review synthesised common CV tasks used in literature including segmentation, object detection, classification, tracking, clustering, predictive modelling, generative synthesis, and immersive visualisation. These tasks are enabled by a wide range of ML and DL models, reflecting the growing technical sophistication of CV-based urban research, as examined in the following subsections. Fig. 8 organises this synthesis as a planning-oriented workflow, progressing from applications to CV task types, associated models and algorithms, and the datasets that support their implementation.

3.3.1. CNN-based approaches in urban analysis

Cutting-edge deep neural network algorithms have demonstrated superior performance in general CV tasks and are increasingly used to extract diverse features of the built environment from street-level, and

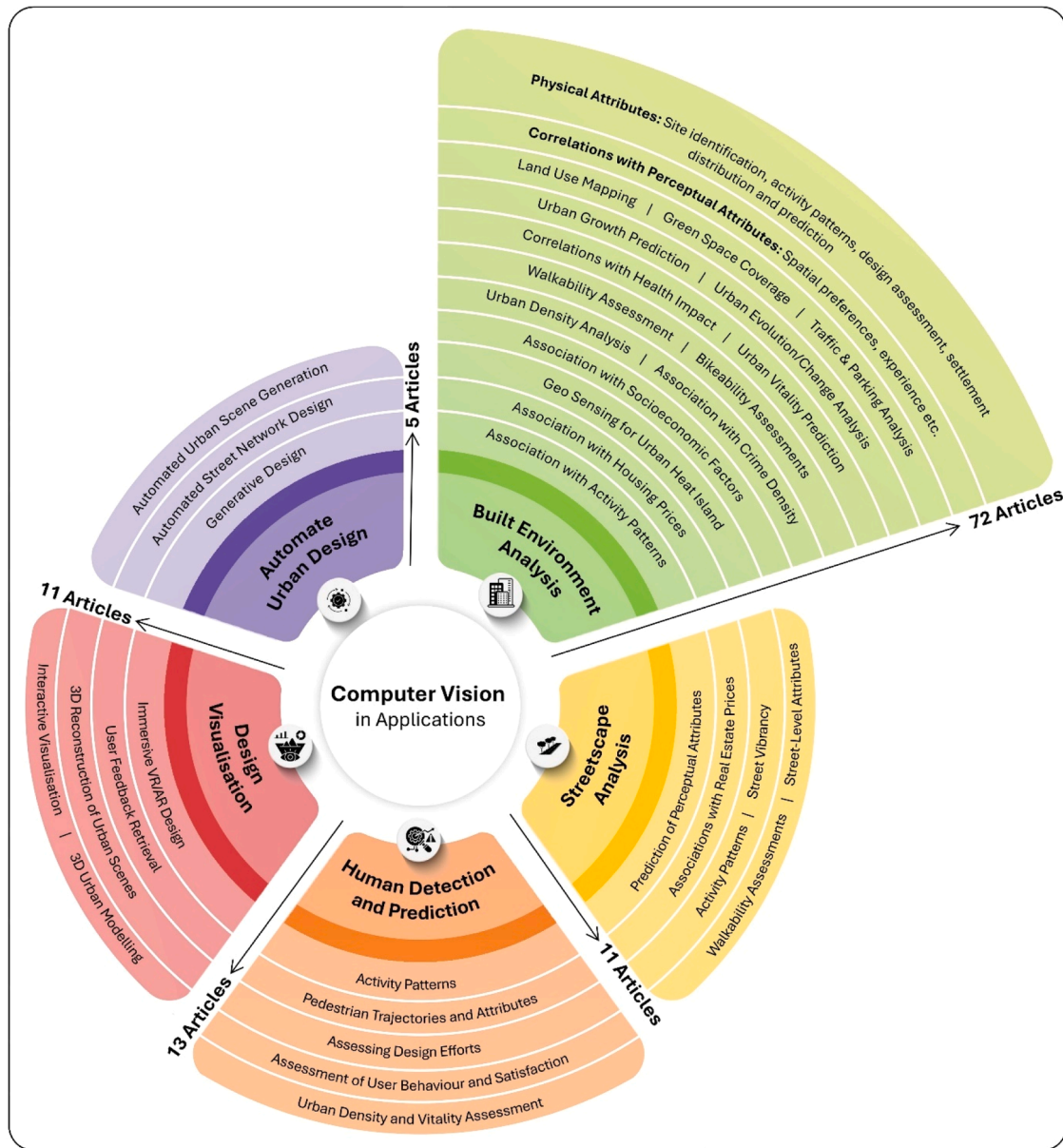


Fig. 6. Detailed categorisation of CV applications in reviewed literature.

satellite imagery. CNNs have long dominated the tasks of image processing and analysis and proven effective in extracting both physical and perceptual features (Verma et al., 2021; Zhao et al., 2023). Across the reviewed literature, CNN-based models are predominantly used for wide range of CV tasks, including image classification, segmentation, and object detection, which together support the systematic measurement of urban form, infrastructure, environmental features, and human presence relevant to planning analysis.

A diverse range of CNN architectures has been adopted in urban studies (Table 4), including region-based CNN (R-CNN) (Koo et al., 2023; Chen & Biljecki, 2023), single shot multibox detector (SSD) (Ibrahim et al., 2021), siamese CNNs (SCNN) (Ilic et al., 2019). These models have demonstrated their effectiveness in detecting and classifying multiple objects with both speed and accuracy (Ibrahim et al., 2021). The success of CNNs as feature extractors can be attributed to their ability to learn image features with high levels of accuracy and performance (Huai et al., 2022). Pre-trained freely accessible models, particularly those trained on benchmark datasets such as Cityscapes, ADE20K, and COCO, are frequently employed to reduce annotation

effort and improve model transferability, although their applicability depends on contextual similarity between training and deployment cities (Tian et al., 2021; Doiron et al., 2022).

Among these architectures, semantic segmentation networks such as PSPNet and DeepLabV3+ are particularly prevalent in urban research, as they assign class labels at the pixel level and enable fine-grained spatial interpretation of streetscapes and built environments (Tian et al., 2021). Pre-trained ResNet backbones are also widely adopted for classification and segmentation due to their practicality and high accuracy in solving complex tasks (Liang et al., 2023), while DeepLabV3+, when trained on the Cityscapes dataset, has demonstrated superior accuracy, outperforming mainstream DL algorithms such as SegNet and PSPNet, making it particularly effective for urban scene understanding (Liang et al., 2023; Zhang et al., 2023). Other commonly used architectures, such as FCN, SegNet, U-Net, and SegFormer, are also widely adopted for detection, segmentation, and classification tasks enabling planners to quantify urban environments at fine spatial resolution (Koo et al., 2022).

Object detection models such as YOLO, VGG-16 and Faster R-CNN

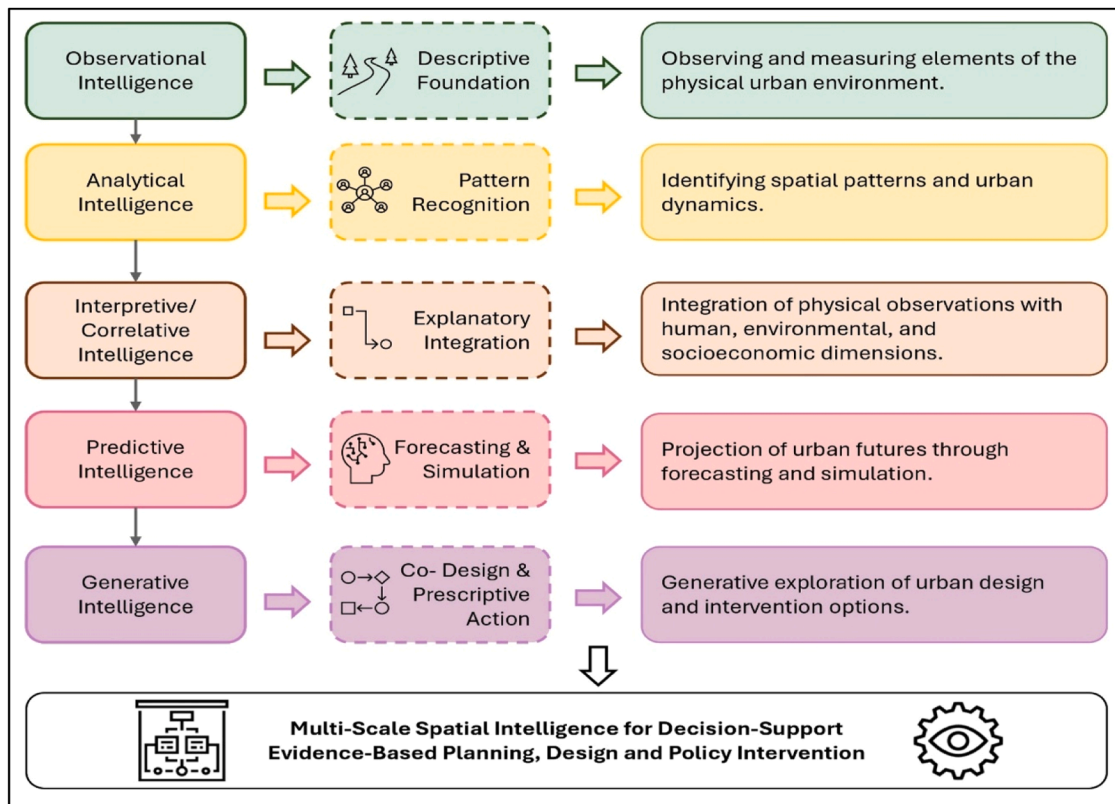


Fig. 7. UVI hierarchy derived from inductive synthesis of reviewed studies.

Table 3
Inductive mapping from reviewed application patterns to the UVI hierarchy.

UVI level	Dominant application patterns observed in reviewed papers	Typical CV tasks	Typical planning function
Observational intelligence	Built environment inventories; land-use mapping; streetscape feature extraction	Segmentation, detection, classification	Baseline audits; monitoring assets and environmental conditions
Analytical intelligence	Activity patterns; mobility flows; spatial and temporal diagnostics	Detection + tracking, segmentation, spatiotemporal analysis	Diagnosing hotspots; urban dynamics and spatial inequality
Correlative intelligence	Linking urban form to perception, health, safety, or socioeconomic outcomes	Feature extraction + regression/classification; explainability methods	Explaining form–outcome relationships; evaluating design performance and experience
Predictive intelligence	Urban growth, impact forecasting, behaviour and risk prediction	ML-based prediction; simulation	Scenario testing; policy appraisal
Generative intelligence	Automated design; text-to-image urban prototyping; immersive visualisation	Generative models; interactive and immersive visualisation	Design exploration and participation

are widely used in mobility and transport studies for real-time detection and counting of pedestrians and vehicles, reflecting their computational efficiency and robustness in dynamic environments. More recent studies increasingly combine CNN backbones with transformer-based components or employ hybrid architectures to improve contextual awareness

and segmentation accuracy in visually complex urban settings (Yi et al., 2023). These advances reflect a shift toward more robust and generalisable feature extraction, particularly for heterogeneous and dense urban environments. Collectively, CNN-based approaches underpin much of the empirical foundation by enabling scalable and consistent spatial feature extraction, with models trained on diverse datasets to enhance performance and generalisability for higher-level analytical, predictive, and design-oriented tasks (Lynda et al., 2021; Ibrahim et al., 2021). These models are valuable for baseline urban audits, environmental monitoring, and spatial diagnostics producing observational and analytical urban intelligence.

3.3.2. Other ML and DL algorithms and their applications

In addition to CNN-based models, a wide range of ML and DL algorithms are employed to support classification, clustering, tracking, prediction, and generative tasks in urban CV applications. Supervised ML algorithms such as random forest classifier (RF) and support vector machine (SVM) are frequently employed for classification and regression tasks, including land-use categorisation, perception estimation, pedestrian volume prediction, environmental quality assessment, and change detection, due to their relative interpretability, accuracy and modest computational requirements (Li et al., 2017; Yenchu, 2019; Sharma et al., 2022).

Recent advances have widely adopted transformer-based DL models (particularly Vision Transformers (ViTs)) as backbone for image classification, detection, and retrieval, thereby expanding their applicability in urban analytics (Jo et al., 2024). ViTs and their variants are increasingly as benchmark against CNN architectures, demonstrating strong performance in urban image classification and other CV tasks (Luo et al., 2022a; Zhao et al., 2023). Recent evidence shows that self-attention architectures increasingly outperform conventional CNNs and non-image-based models, underscoring the growing relevance of transformer approaches in urban CV research (Mao et al., 2022a; Zhao et al., 2023).

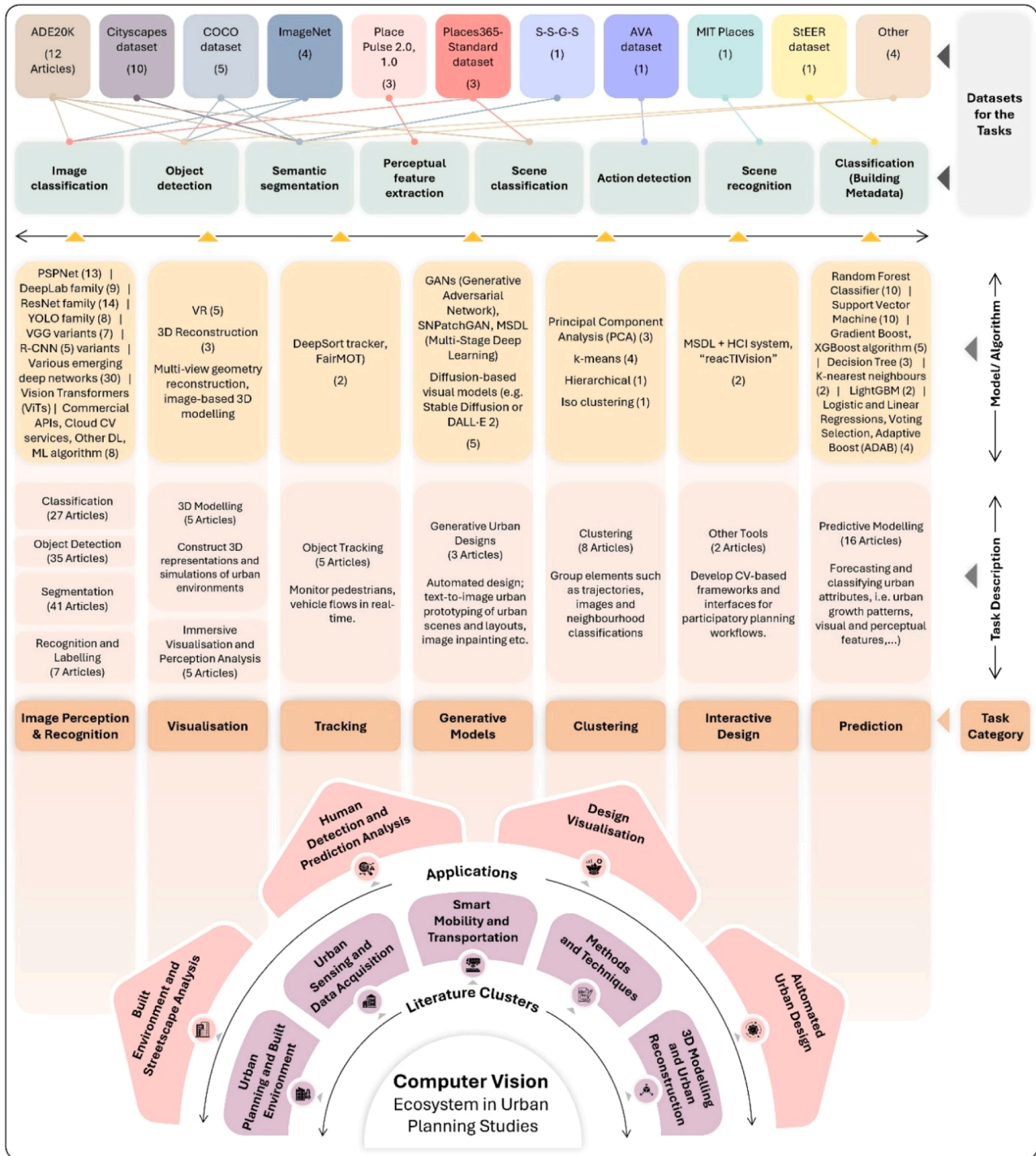


Fig. 8. Computer vision ecosystem in urban planning research.

Rapid advances in foundational models, including vision-based systems, are reshaping smart city research by enabling highly generalisable, zero-shot systems, from segmentation models like Segment Anything to language-augmented models such as CLIP for urban scene interpretation (Zhang et al., 2025). While task-agnostic models show strong performance in text-based geospatial tasks, evidence suggests they still lag behind task-specific models when integrating multimodal geospatial data (Mai et al., 2024). Nevertheless, emerging urban foundation models (e.g. large ViTs and self-supervised networks) and multimodal CV systems are redefining visual analysis in urban contexts supporting diverse planning tasks; from land-use classification, semantic

segmentation to urban inquiries and infrastructure documentation (Zhang et al., 2025).

Clustering techniques (including k-means clustering and principal component analysis) are commonly applied to explore spatial patterns and thematic structures in CV-derived feature spaces (Wang et al., 2022; Liang et al., 2023). These approaches are particularly useful for exploratory analysis of urban form, spatial preferences, and neighbourhood typologies. For video-based applications, tracking algorithms such as DeepSort and FairMOT are often integrated with object detection and clustering models to analyse pedestrian and vehicle trajectories, supporting studies of mobility flows, crowd dynamics, and

Table 4
Commonly used models and algorithms for CV-based urban research.

Task category	Models/algorithms	Task description & planning application	No. of reviewed papers	Exemplar study
Image Perception & Recognition	PSPNet	Performs semantic segmentation, image classification, object detection, and scene recognition to identify, classify, and generate semantic information about urban elements (e.g., vehicles, scenes, land uses)	13	(Koo et al., 2023)
	DeepLab family		9	(Chen & Biljecki, 2023)
	ResNet family		14	(Stalder et al., 2024)
	YOLO family		8	(Salazar-Miranda et al., 2023)
	VGG variants	Facial recognition, traffic analysis, segmentation and image labelling	7	(Suel et al., 2023)
	R-CNN variants		5	(Koo et al., 2023)
	Various emerging deep networks (ConvNeXt, EfficientDet, U-Net, DeepForest, RetinaNet, HRNet, SlumsNet, AlexNet, ViTs; Swin, SegFormer)		30	(Ibrahim et al., 2021; Luo et al., 2022b; Zhang et al., 2025)
	Commercial APIs (e.g., FacePlusPlus, GoodVision)		8	(O'Regan et al., 2021; Song et al., 2022b; Ghermandi et al., 2022)
	Cloud CV services (e.g. Clarifai, Google Vision and Microsoft Azure AI Vision)			
	Other DL, ML algorithms (Pymeanshift, ACAM)			
Tracking	DeepSort tracker, FairMOT	Monitor pedestrian or vehicle flows in real-time through object tracking	2	(Wong et al., 2021; Niu et al., 2022)
Visualisation	VR	Simulates urban environments for interactive perception and real-time feedback	5	(Sun et al., 2020; Yuan et al., 2023)
	3D Model Reconstruction	3D city model building	3	(Zou et al., 2012; Verma et al., 2021)
Clustering	“Principal Component Analysis (PCA)”	Grouping spatial data such as trajectories and classifications	3	(Wang et al., 2022; Niu et al., 2022)
	“k-means”		4	
	“Hierarchical”		1	
Generative Models	“Iso clustering”	Generative urban designs—i.e. urban scene beautification, street network generation, image inpainting	1	(Fang et al., 2022; Joglekar et al., 2020; Valença et al., 2025)
	GANs, SNPatchGAN, MSDL (Multi-Stage Deep Learning)		5	
	Emerging technologies such as diffusion-based visual models (e.g. Stable diffusion or DALL-E 2)			
Interactive Design	MSDL + HCI system, “reacTIVision”	Human–computer interaction frameworks and interfaces for participatory urban design	2	(Fang et al., 2022; Maquil et al., 2018)
Prediction	RF Classifier	Prediction and classification of urban attributes	10	(Sharma et al., 2022; Su et al., 2023)
	SVM		10	(Xu et al., 2022; Lee et al., 2022)
	Gradient Boost, XGBoost algorithm		5	
	Decision tree		3	
	K-nearest neighbours		2	
	LightGBM		2	
	Logistic and linear regressions, Voting selection, Adaptive Boost (ADAB)		4	

public-space usage (Wong et al., 2021; Niu et al., 2022). These approaches provide fine-grained evidence on urban vitality, accessibility, and safety, and are particularly valuable for identifying latent spatial structures and informing diagnostic planning tasks.

For predictive tasks, CV-derived features are frequently combined with supervised ML models such as RF classifier, SVM and LightGBM to model outcomes related to urban perception, mobility patterns, and urban growth dynamics due to their high reliability and minimal computing cost in training (Ito & Biljecki, 2021; Jaad & Abdelghany, 2021; Salazar-Miranda et al., 2023). Studies often compare several classic ML and DL algorithms, such as lasso, ridge, SVM, k-nearest neighbour, deep neural networks, and tree-based algorithms like RF, GBDT, LightGBM, XGBoost, along with logistic regression, to identify the best performing algorithms for prediction (Li & Ma, 2022). In some cases, explainable AI techniques, such as SHAP, are used to improve transparency and interpretability of model outputs, particularly where results inform planning evaluation or policy analysis (Lee et al., 2022). Predictive modelling represents a key step in increasing the decision-relevance of CV outputs.

Generative models represent a rapidly growing strand of CV research in urban planning. Studies employ GANs, multi-stage DL frameworks, and diffusion-based models to support automated urban design exploration, streetscape enhancement, image inpainting, visualisation tasks,

and scenario prototyping (Joglekar et al., 2020; Fang et al., 2022; Valença et al., 2025). Advances in generative AI expand urban design capabilities by enabling context-specific layouts, interactive human–computer–integrated exploration and the generation of realistic and aesthetically coherent designs (Fang et al., 2022; Wang et al., 2025). While these approaches enable rapid visual exploration and participatory engagement, their application remains limited in practice and raises important challenges related to contextual validity, functional feasibility, and bias—issues that warrant careful consideration as generative methods become more integrated into planning workflows.

Diffusion-based visual and text-to-image models (e.g. Stable Diffusion and DALL-E 2) have become central to DL-driven image generation, enabling high-fidelity urban design prototyping, iterative refinement and rapid visualisation that support conceptual designs and participatory planning approaches (Jo et al., 2024; Ehab et al., 2025). Recent studies demonstrate the potential of diffusion- and transformer-based generative models for urban analytics and design, including foundation models for open-world urban spatiotemporal learning, text-to-image generation of realistic urban and satellite imagery, and planning-relevant built-environment visualisation without domain-specific fine-tuning (Seneviratne et al., 2022; Yuan et al., 2025; Wang et al., 2025). These models show robust, adaptable performance across urban tasks and cities, with strong alignment to planning

constraints.

CV-enabled visualisation—often combined with virtual or augmented reality—allows planners and stakeholders to explore urban environments from human-centred perspectives, compare design alternatives, and evaluate perceptual and experiential impacts of proposed interventions (Sun et al., 2020; Yuan et al., 2023). 3D model reconstruction techniques are used to generate spatial models that support effective visualisation and analysis of urban environments (Irschara et al., 2012; Landeschi et al., 2020). Interactive systems incorporating human–computer interaction frameworks and tangible user interfaces enable real-time feedback, collaborative exploration, and iterative refinement of design options (Maquil et al., 2018; Fang et al., 2022). These applications play a crucial role in operationalising generative intelligence, where CV outputs move beyond analysis to support co-creative planning and participatory engagement (D’Amico et al., 2020; Yuan et al., 2023).

Alongside bespoke academic models, commercial CV platforms and cloud-based APIs have become increasingly prominent in applied urban planning contexts (Huai et al., 2022; Ghermandi et al., 2022). Platforms such as Google Cloud Vision, Microsoft Azure AI Vision, Clarifai, and GoodVision provide access to pretrained models for image labelling, object detection, and traffic analysis, lowering technical barriers for practitioners and enabling rapid deployment at scale. While these tools facilitate exploratory analysis and monitoring, their use introduces challenges related to model transparency, contextual validity, and reproducibility. As such systems operate as black boxes with limited control over training data and inference logic, their outputs must be interpreted cautiously and supplemented with domain expertise, particularly when informing planning assessments or public decision-making.

Collectively, the integration of diverse methods within planning workflows determines whether CV outputs remain descriptive or evolve into actionable decision-support tools capable of informing evaluation, scenario testing, and intervention design across increasing levels of decision relevance.

3.4. CV training datasets

Training datasets are critical for CV-based urban analysis. The reviewed studies rely heavily on publicly available benchmark datasets, as well as manually annotated, context-specific datasets (Table 5). While widely used benchmarks facilitate comparability and rapid model development, their geographic concentration and original design purposes limit their representativeness of diverse urban contexts. Several studies address this limitation by supplementing benchmark datasets with locally annotated imagery or by fine-tuning models on context-specific data to improve applicability (Verma et al., 2021).

Multilayer CNN-based models require large volumes of training data, leading urban CV research to rely heavily on established benchmark datasets. For example, models pre-trained on ADE20K (a large-scale scene understanding dataset with 150 segmentation classes) have been widely used because they offer broader scene coverage than more narrowly scoped datasets such as Cityscapes (Zhang et al., 2023). Other widely used resources include COCO, which supports object detection across 80 object classes (Knura et al., 2021; Song et al., 2022a), Cityscapes, which is tailored to semantic understanding of urban street environments with 30 object categories (Liu et al., 2023), and ImageNet, a foundational dataset that underpins many modern CNN and DL models (Matsuzaka & Yashiro, 2023; Vanky & Le, 2023).

Domain-specific datasets are also employed, such as S-S-G-S for urban vegetation classification (Zhang et al., 2023) and Place Pulse for analysing urban landscape perception using globally sourced imagery (Luo et al., 2022b). In addition, specialised datasets targeting human actions and facial expressions are used where relevant, with commonly adopted training datasets summarised in Table 5. The reviewed datasets reveal both the strengths and constraints of current CV-based urban

Table 5

Commonly used training datasets.

Dataset	Task	Description	No. of studies	Exemplar study
ADE20K	Image classification, Object detection, Semantic segmentation, Scene parsing	A large, annotated dataset containing over 20,000 scene-centric images with pixel-level labels across >3000 object categories. Includes complex everyday scenes in natural spatial context, widely used for scene understanding and urban analysis.	12	(Zhou et al., 2019; Scepianovic et al., 2021; Suel et al., 2023)
Cityscapes dataset	Semantic segmentation, Instance segmentation	High-resolution street-level imagery from 50 European cities with detailed pixel-level annotations, designed for urban scene understanding and autonomous driving research.	10	(Zhang et al., 2023; Liu et al., 2023)
COCO dataset	Object detection, Segmentation	Over 200,000 images with rich annotations for objects in natural context, enabling complex urban-environment recognition tasks.	5	(Salazar-Miranda et al., 2023)
ImageNet	Image classification, Object detection	Contains 14 million annotated images across 20,000 categories; foundational for training CNNs and DL models, including AlexNet, VGG, and ResNet.	4	(Deng et al., 2009; Stalder et al., 2024)
Place Pulse 2.0, 1.0	Perceptual feature extraction	Street-view image dataset annotated through crowdsourced pairwise comparisons to infer urban perception attributes such as safety and liveliness.	3	(Liang et al., 2023)
Places365-Standard dataset	Image, scene classification	Contains 1.8 million training images across 365 scene	3	(Zhou et al., 2018; Chen & Biljecki, 2023)

(continued on next page)

Table 5 (continued)

Dataset	Task	Description	No. of studies	Exemplar study
S-S-G-S	Semantic segmentation	categories; widely used for large-scale scene recognition related to urban built environments. Street view dataset mainly used for the analysis of urban street vegetation communities with high accuracy.	1	(Zhang et al., 2023)
AVA dataset	Action detection	A large-scale video dataset annotated for human actions and events; used to analyse human activities and behaviours in public urban spaces.	1	(Zhang et al., 2022)
MIT Places	Scene recognition	An earlier large-scale scene dataset containing millions of labelled images for scene understanding.	1	(Payntar, 2022)
StEER dataset	Classification (Building Metadata)	A dataset created for classifying building types and extracting structural information from street-level imagery for urban analysis.	1	(Liu et al., 2024)
Other	Segmentation, object detection	Diverse datasets are used such as Mapillary Vistas with many infrastructure-related categories for bikeability assessment and manually created training datasets for urban assessments.	4	(Ito & Biljecki, 2021; Verma et al., 2021)

research. Continued expansion of inclusive, planning-oriented datasets and improved documentation of data provenance and bias will be essential to support fair, transferable, and context-sensitive urban analysis.

4. Findings and discussion

4.1. Observational and analytical intelligence: advancing urban evidence and diagnosis

At the foundational levels of UVI (observational and analytical

intelligence) the reviewed literature demonstrates the strongest maturity. A large proportion of studies focus on extracting measurable features of the built environment and diagnosing spatial and temporal patterns using CV techniques. The identified research clusters related to urban planning, sensing, mobility, and methods reflect this emphasis on large-scale measurement, monitoring, and diagnostic analytics.

CV-derived inventories of land use, streetscape elements, environmental conditions, and mobility flows address long-standing evidence gaps in urban planning, where data collection has traditionally been resource-intensive, infrequent, and spatially coarse. The widespread use of segmentation, detection, and tracking enables planners to move from static representations toward dynamic, fine-grained diagnostics of urban systems. The integration of CV, particularly through DL and neural network architectures, has increasingly shaped urban analytics. The prominence of land-use mapping and mobility focused studies reflects the effectiveness and accuracy of DL-based classification and tracking for urban monitoring (Mao et al., 2022b; Zhang & Berger, 2023).

The emergence of 3D reconstruction as a research cluster corresponds with the growing adoption of digital twins and urban modelling platforms, reflecting a shift toward analytical intelligence that enables detailed diagnosis, simulation, and evaluation of existing urban systems (Lei & Biljecki, 2025). These analytical representations increasingly serve as foundational inputs for predictive and generative applications, including scenario testing and design exploration.

These findings suggest that CV has already become a reliable instrument for baseline assessment and problem diagnosis, particularly in areas such as mobility management, environmental monitoring, and built-environment evaluation. It supports real-time data collection, interpretive decision support, allowing researchers and planners to better address complex urban challenges. However, automated visual analysis may also oversimplify urban experience by privileging surface-level visual cues over broader social and cultural contexts (Vanky & Le, 2023). Also, the concentration of studies at these intelligence levels also highlights an imbalance: much of the literature stops at measurement and pattern detection, with fewer studies translating diagnostics into explanatory or forward-looking planning insights.

4.2. Correlative intelligence: explaining urban form-outcome relationships

At the level of correlative intelligence, CV enables planners to empirically examine relationships between urban form and social, perceptual, environmental, and economic outcomes. The review reveals a growing body of work linking CV-extracted features to perceived safety, walkability, health outcomes, crime, and property values, addressing a persistent explanatory gap in planning research (Naik et al., 2017; Gebru et al., 2017).

By operationalising visual characteristics at scale, CV allows long-standing planning hypotheses (such as the relationship between street-scape design and perceived safety) to be tested quantitatively across large urban areas. This represents a substantive advance beyond descriptive mapping by supporting evidence-based, people-centred evaluation of design performance while enabling deeper analysis of human–environment interactions.

Nevertheless, limitations remain. Correlative intelligence is highly sensitive to data bias, contextual misinterpretation, and proxy validity. Many studies rely on imagery from a limited set of cities, raising concerns about generalisability and equity when findings are used to inform planning decisions in diverse contexts. These issues underscore the need for stronger validation, interpretability, and context-awareness before correlative insights are embedded in policy.

4.3. Predictive intelligence: scenario testing and planning risk

Predictive intelligence represents a shift from explaining existing conditions to anticipating future urban dynamics. The review shows increasing use of CV-derived variables in forecasting urban growth,

mobility demand, environmental exposure, and behavioural patterns. This capacity enhances scenario testing and policy appraisal, particularly where traditional forecasting models are constrained by limited or outdated inputs.

However, predictive intelligence introduces higher stakes. Errors, biases, or overconfident predictions can propagate directly into planning decisions, resource allocation, and long-term urban outcomes. The review highlights persistent challenges related to model transferability, uncertainty reporting, and explainability, which limit the safe use of predictive CV outputs in planning practice. The black-box nature of DL models undermines trust and constrains the use of CV insights in public decision-making (Buhrmester et al., 2021).

These findings reinforce the importance of treating predictive CV as decision-support rather than decision-making tools, requiring explicit uncertainty communication and governance safeguards.

4.4. Generative intelligence: Design exploration, hallucination risks, and planning validity

Generative intelligence represents a rapidly evolving level of UVI, with growing potential but comparatively limited empirical validation in real-world planning contexts. Emerging applications of generative CV (such as GAN- and diffusion-based urban scene synthesis, automated street network generation, and text-to-image design prototyping) offer promising avenues for rapid design exploration and participatory visualisation (Jo et al., 2024; Valença et al., 2025). These AI-assisted creative processes support scenario-driven urban planning, enhance design efficiency, and expand the solution space in early-stage planning (He et al., 2026).

Nonetheless, the review also reveals a critical risk that has received limited attention in the planning literature: hallucination. Generative models can produce urban forms that are visually photorealistic yet functionally invalid, structurally infeasible, or incompatible with planning regulations, engineering constraints, or local context. Such hallucinations are particularly problematic in planning, where visual plausibility may mask violations of accessibility standards, infrastructure capacity, safety requirements, or land-use controls. Automated generative design may also fail to capture the tacit and contextual judgement of human planners (Valença et al., 2025).

Unlike analytical CV tasks, generative outputs are not constrained by empirical ground truth. Without explicit conditioning on planning rules, physical constraints, or domain knowledge, generative models may synthesise urban structures that appear coherent but would fail in real-world implementation. This creates a risk of aesthetic bias, where visually compelling outputs disproportionately influence decision-making despite limited functional validity.

Recent work has begun to address these challenges; for example, Wang et al. (2025) adapt a Stable Diffusion–ControlNet framework using widely available geospatial data to produce more realistic and context-sensitive urban imagery. However, issues of scalability, access to labelled urban data, and practical reliability persist, underscoring the need for systematic evaluation of prompt and control-image effects and qualitative validation to ensure generative outputs reflect real-world design constraints rather than reinforcing hidden biases or modelling artefacts.

The findings therefore caution against uncritical adoption of generative CV in urban planning. Generative intelligence should be framed as exploratory and supportive, not authoritative. Robust integration requires constraint-aware generation, human-in-the-loop validation, and explicit differentiation between conceptual visualisation and implementable design proposals. These concerns underscore the need to evaluate CV-generated outputs against established planning practices and values.

4.5. Cross-Cutting limitations and governance across the UVI hierarchy

Across all UVI levels, recurring challenges related to bias, interpretability, privacy, and governance persist. As intelligence levels increase, so do the consequences of error and misuse. While observational errors may affect monitoring accuracy, errors at predictive or generative levels can shape long-term urban form and social outcomes.

For instance, models trained on visually distinct cities may not generalise well to other regions, raising concerns about dataset representativeness (Suel et al., 2023; Ito et al., 2024). Data quality issues, noise, and inaccuracies in feature extraction (such as misclassification and biases from uneven data or inaccurate locational inputs) can skew results, undermine fairness, explainability, and overall analytical reliability (Huai et al., 2022; Chen & Biljecki, 2023). At the same time, strategies such as model tailoring, data diversification, and augmentation are used to improve robustness (Chen & Biljecki, 2023).

The findings highlight not only the maturity of CV applications in core urban domains but also gaps in underexplored areas such as participatory planning, and governance. The effectiveness of CV methods compared to traditional practices must be critically evaluated, particularly in terms of interpretability and implications for urban decision-making.

Furthermore, the true transformative potential of CV lies in integrating these disparate analyses into a unified, multi-scale diagnostic framework for urban planning. The review highlights the need for explainable AI, diverse and context appropriate datasets, and governance frameworks that align CV deployment with planning values to ensure responsible AI application in urban planning (Kang et al., 2026). Importantly, limitations should not be treated as secondary considerations but as design constraints that shape responsible adoption.

4.6. Future directions: From visual intelligence to planning intelligence

Future research should prioritise balanced development across the UVI hierarchy. While observational and analytical intelligence are well established, greater attention is needed on validating correlative, predictive, and generative applications in real planning contexts.

Advances in design visualisation and automation enable more precise representations of urban spaces and enhance spatial comprehension through immersive, participatory design experimentation (Globa et al., 2022; Zhao, 2024). The growing accessibility of commercial CV platforms, pre-trained DL models, and publicly available datasets has enabled urban researchers and planners to adopt advanced visual analytics without extensive technical expertise, supporting broader innovation in urban studies (Ghermandi et al., 2022). State-of-the-art CV methods are widely applied in urban research through CNN-based and increasingly transformer-based architectures, often combined with ML techniques, to enhance the robustness, accuracy and contextual understanding in complex urban environments (Zhang & Berger, 2023; Yi et al., 2023; Stalder et al., 2024).

Emerging diffusion-based generative models enabling high-fidelity design visualisation, participatory planning, and scenario testing (Jo et al., 2024). ViTs and foundation models (i.e. Segment Anything, CLIP, and DINO) supporting scalable, zero-shot, generalisable, rapid adaptation across datasets, and multi-modal urban analysis (Jo et al., 2024; Zhao et al., 2023; Zhou et al., 2024). Together, these advances point toward more adaptive, data-rich, and inclusive urban analytics, though effective adoption will require domain-specific tailoring, rigorous validation, integration with planning expertise, and careful attention to social, cultural, and ethical considerations.

Advancing urban visual intelligence into planning intelligence requires interdisciplinary collaboration, domain-specific foundation models, and stronger integration of legal, ethical, and institutional constraints. Only through such integration can CV move from technical innovation to sustainable urban transformation.

5. Conclusion

This review synthesises the rapidly expanding literature on CV in urban planning, examining how contemporary models, methods, and data sources are reshaping traditional approaches to urban analysis, monitoring, and design. The paper identified five dominant research clusters (urban planning and built environment analysis, urban sensing and data acquisition, smart mobility, methods and techniques, and 3D urban modelling) highlighting the breadth and interdisciplinary nature of CV applications in urban studies. Across these domains, CV enables the extraction of spatially explicit, fine-grained evidence on urban form, activity, and infrastructure that would otherwise be costly, slow, or impractical to obtain using conventional planning tools.

Beyond cataloguing applications, the review advances three core insights. First, CV has matured into a foundational analytical infrastructure for urban planning, supporting a wide range of functions from baseline measurement and diagnostics to performance evaluation and monitoring. Second, advances in deep learning (initially driven by convolutional neural networks and increasingly shaped by vision transformers, diffusion models, and emerging urban foundation models) has substantially increased the accuracy, scalability, and adaptability of CV-based urban analytics. Third, CV is moving beyond descriptive and predictive analysis toward co-creative and participatory design support, enabling interactive visualisation, rapid scenario exploration, and early-stage design prototyping that can meaningfully engage both experts and communities.

The UVI framework proposed in this study integrates these developments into a coherent, decision-oriented structure. By conceptualising CV as a progression from observational and analytical intelligence to correlative, predictive, and generative intelligence, the framework clarifies how visual data can be transformed into planning-relevant knowledge with increasing decision impact. Importantly, the framework also highlights that each level of intelligence carries risks, reinforcing the need for stronger validation, interpretability, and governance as CV outputs move closer to policy and design action. In this way, the review reframes CV not merely as a collection of technical methods, but as a planning-intelligence system that must be aligned with institutional responsibilities and public accountability.

The findings have clear implications for planning practice and policy. CV can strengthen evidence-based decision-making, enhance the monitoring of urban conditions, and support more adaptive and responsive planning interventions. It also offers new opportunities to incorporate perceptual and experiential dimensions—such as safety, comfort, and environmental quality—into planning evaluations, thereby improving sensitivity to lived urban realities. At a governance level, CV-enabled systems can enhance transparency, streamline routine assessments, and broaden public engagement through accessible visual and interactive tools.

Nevertheless, the benefits of CV will not be realised without addressing persistent challenges. Dataset biases, limited transferability across urban contexts, and the opacity of DL models raise concerns regarding fairness, accountability, and trust. In the case of generative CV, additional risks arise from hallucinations; where visually convincing outputs may be functionally invalid, structurally infeasible, or misaligned with regulatory and infrastructural constraints. Ethical risks—including privacy breaches, unintended surveillance, and algorithmic discrimination. These risks highlight the importance of treating CV, particularly at predictive and generative levels, as decision support rather than decision authority, supported by explainable AI, human oversight, and robust governance frameworks.

Looking ahead, CV holds substantial promise for enabling more adaptive, inclusive, and sustainable urban futures. Realising this potential requires advances not only in algorithms, but also in planning-oriented validation, domain-adaptable and multimodal urban foundation models, and human–AI collaborative platforms that support co-creative and participatory planning. Strengthening interdisciplinary

collaboration, embedding ethical safeguards, and aligning CV deployment with planning values and institutional contexts will be essential. When integrated strategically and responsibly, CV can serve as a powerful enabler of smarter, more resilient, and more equitable cities.

Funding

This research was funded by the Australian Research Council Discovery Grant Scheme, grant number DP220101255.

CRediT authorship contribution statement

Raveena Marasinghe: Writing – original draft, Formal analysis, Data curation. **Tan Yigitcanlar:** Writing – review & editing, Supervision, Conceptualization. **Xinyu Fu:** Writing – review & editing. **Steven Jige Quan:** Writing – review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgments

The authors thank the editor and anonymous referees for their constructive comments on an earlier version of the manuscript.

Data availability

Data will be made available on request.

References

- Azfar, T., Li, J., Yu, H., Cheu, R. L., Lv, Y., & Ke, R. (2024). Deep learning-based computer vision methods for complex traffic environments perception: A review. *Data Science for Transportation*, 6(1), 1. <https://doi.org/10.1007/s42421-023-00086-7>
- Biljecki, F. (2023). GeoAI for urban sensing. *Handbook of Geospatial Artificial Intelligence*. CRC Press.
- Buhrmester, V., Münch, D., & Arens, M. (2021). Analysis of explainers of black box deep neural networks for computer vision: A survey. *Machine Learning and Knowledge Extraction*, 3(4), 4. <https://doi.org/10.3390/make3040048>
- Chen, S., & Biljecki, F. (2023). Automatic assessment of public open spaces using street view imagery. *Cities (London, England)*, 137, Article 104329. <https://doi.org/10.1016/j.cities.2023.104329>
- D'Amico, G., L'Abbate, P., Liao, W., Yigitcanlar, T., & Ioppolo, G. (2020). Understanding sensor cities: Insights from technology giant company driven smart urbanism practices. *Sensors*, 20(16), 16. <https://doi.org/10.3390/s20164391>
- Doiron, D., Setton, E., Brook, J., Kestens, Y., McCormack, G., Winters, M., Shooshtari, M., Azami, S., & Fuller, D. (2022). Predicting walking-to-work using street-level imagery and deep learning in seven Canadian cities. *Scientific Reports*, 12(1), 1. <https://doi.org/10.1038/s41598-022-22630-1>
- Ehab, A., Aladawi, A., & Burnett, G. (2025). Exploring AI-integrated VR systems: A methodological approach to inclusive digital urban design. *Urban Science*, 9(6), 6. <https://doi.org/10.3390/urbansci9060196>
- Faisal, A., Yigitcanlar, T., Kamruzzaman, M., & Paz, A. (2021). Mapping two decades of autonomous vehicle research: A systematic scientometric analysis. *Journal of Urban Technology*, 28(3–4), 45–74. <https://doi.org/10.1080/10630732.2020.1780868>
- Fang, Z., Qi, J., Fan, L., Huang, J., Jin, Y., & Yang, T. (2022). A framework for human-computer interactive street network design based on a multi-stage deep learning approach. *Computers, Environment and Urban Systems*, 96, Article 101853. <https://doi.org/10.1016/j.compenvurbsys.2022.101853>
- Gebri, T., Krause, J., Wang, Y., Chen, D., Deng, J., Aiden, E. L., & Fei-Fei, L. (2017). Using deep learning and Google Street View to estimate the demographic makeup of neighborhoods across the United States. *Proceedings of the National Academy of Sciences*, 114(50), 13108–13113. <https://doi.org/10.1073/pnas.1700035114>
- Ghermandi, A., Depietri, Y., & Sinclair, M. (2022). In the AI of the beholder: A comparative analysis of computer vision-assisted characterizations of human-nature interactions in urban green spaces. *Landscape and Urban Planning*, 217, Article 104261. <https://doi.org/10.1016/j.landurbplan.2021.104261>
- Globa, A., Beza, B. B., & Wang, R. (2022). Towards multi-sensory design: Placemaking through immersive environments – Evaluation of the approach. *Expert Systems with Applications*, 204, Article 117614. <https://doi.org/10.1016/j.eswa.2022.117614>
- Guzder-Williams, B., Mackres, E., Angel, S., Blei, A. M., & Lamson-Hall, P. (2023). Intra-urban land use maps for a global sample of cities from Sentinel-2 satellite imagery and computer vision. *Computers, Environment and Urban Systems*, 100, Article 101917. <https://doi.org/10.1016/j.compenvurbsys.2022.101917>

- He, M., Liang, Y., Wang, S., Zheng, Y., Wang, Q., Zhuang, D., Tian, L., & Zhao, J. (2026). Human-guided urban form generation using multimodal diffusion models. *Building and Environment*, 287, Article 113892. <https://doi.org/10.1016/j.buildenv.2025.113892>
- Huai, S., Chen, F., Liu, S., Canters, F., & Van de Voorde, T. (2022). Using social media photos and computer vision to assess cultural ecosystem services and landscape features in urban parks. *Ecosystem Services*, 57, Article 101475. <https://doi.org/10.1016/j.ecoser.2022.101475>
- Ibrahim, M. R., Haworth, J., & Cheng, T. (2021). URBAN-i: From urban scenes to mapping slums, transport modes, and pedestrians in cities using deep learning and computer vision. *Environment and Planning B: Urban Analytics and City Science*, 48(1), 76–93. <https://doi.org/10.1177/2399808319846517>
- Ilic, L., Sawada, M., & Zarzelli, A. (2019). Deep mapping gentrification in a large Canadian city using deep learning and Google Street View. *Plos One*, 14(3), Article e0212814. <https://doi.org/10.1371/journal.pone.0212814>
- Irschara, A., Zach, C., Klopschitz, M., & Bischof, H. (2012). Large-scale, dense city reconstruction from user-contributed photos. *Computer Vision and Image Understanding*, 116(1), 2–15. <https://doi.org/10.1016/j.cviu.2011.07.010>
- Ito, K., & Biljecki, F. (2021). Assessing bikeability with street view imagery and computer vision. *Transportation Research Part C: Emerging Technologies*, 132, Article 103371. <https://doi.org/10.1016/j.trc.2021.103371>
- Ito, K., Quintana, M., Han, X., Zimmermann, R., & Biljecki, F. (2024). Translating street view imagery to correct perspectives to enhance bikeability and walkability studies. *International Journal of Geographical Information Science*, 38(12), 2514–2544. <https://doi.org/10.1080/13658816.2024.2391969>
- Jaad, A., & Abdelghany, K. (2021). The story of five MENA cities: Urban growth prediction modeling using remote sensing and video analytics. *Cities (London, England)*, 118, Article 103393. <https://doi.org/10.1016/j.cities.2021.103393>
- Jo, H., Lee, J., Lee, Y., & Choo, S. (2024). Generative artificial intelligence and building design: Early photorealistic render visualization of façades using local identity-trained models. *Journal of Computational Design and Engineering*, 11(2), 85–105. <https://doi.org/10.1093/jcde/qwae017>
- Joglekar, S., Quercia, D., Redi, M., Aiello, L., Kauer, T., & Sastry, N. (2020). FaceLift: A transparent deep learning framework to beautify urban scenes. *Royal Society Open Science*, 7(1), Article 190987. <https://doi.org/10.1098/rsos.190987>
- Kang, Y., Chen, J., Liu, L., Sharma, K., Mazzarello, M., Mora, S., Duarte, F., & Ratti, C. (2026). Decoding human safety perception with eye-tracking systems, street view images, and explainable AI. *Computers, Environment and Urban Systems*, 123, Article 102356. <https://doi.org/10.1016/j.compenvurbysys.2025.102356>
- Knura, M., Kluger, F., Zahitla, M., Schiewe, J., Rosenhahn, B., & Burghardt, D. (2021). Using object detection on social media images for urban bicycle infrastructure planning: A case study of Dresden. *ISPRS International Journal of Geo-Information*, 10(11), 11. <https://doi.org/10.3390/ijgi10110733>
- Koch, D., Despotovic, M., Thaler, S., & Zeppelzauer, M. (2021). Where do university graduates live? A computer vision approach using satellite images. *Applied Intelligence*, 51(11), 8088–8105. <https://doi.org/10.1007/s10489-021-02268-8>
- Koo, B., Guhathakurta, S., & Botchwey, N. (2022). How are neighborhood and street-level walkability factors associated with walking behaviors? A big data approach using street view images. *Environment and Behavior*, 54(1), 211–241. <https://doi.org/10.1177/00139165211014609>
- Koo, B., Guhathakurta, S., Botchwey, N., & Hipp, A. (2023). Can good microscale pedestrian streetscapes enhance the benefits of macroscale accessible urban form? An automated audit approach using Google street view images. *Landscape and Urban Planning*, 237, Article 104816. <https://doi.org/10.1016/j.landurbplan.2023.104816>
- Landeschi, G., Lindgren, S., Gerding, H., Papadimitriou, A., & Wallensten, J. (2020). Ancient Hermione revealed: The contribution of high-performance computing and digital methods to the analysis of a hidden cityscape. *Archaeological Prospection*, 27(4), 315–328. <https://doi.org/10.1002/arp.1775>
- Lee, J., Kim, D., & Park, J. (2022). A machine learning and computer vision study of the environmental characteristics of streetscapes that affect pedestrian satisfaction. *Sustainability*, 14(9), 9. <https://doi.org/10.3390/su14095730>
- Lei, B., & Biljecki, F. (2025). Evaluating the development of open 3D city models: A multidimensional assessment. *ISPRS Annals of the photogrammetry*. In *Remote Sensing and Spatial Information Sciences, X-4-W6-2025*, 137–144. 20th 3D GeoInfo Conference - 2–5 September 2025. <https://doi.org/10.5194/isprs-annals-X-4-W6-2025-137-2025>
- Li, Z., & Ma, J. (2022). Discussing street tree planning based on pedestrian volume using machine learning and computer vision. *Building and Environment*, 219, Article 109178. <https://doi.org/10.1016/j.buildenv.2022.109178>
- Li, X., Zhang, C., & Li, W. (2017). Building block level urban land-use information retrieval based on Google Street View images. *GIScience & Remote Sensing*, 54(6), 819–835. <https://doi.org/10.1080/15481603.2017.1338389>
- Liang, X., Zhao, T., & Biljecki, F. (2023). Revealing spatio-temporal evolution of urban visual environments with street view imagery. *Landscape and Urban Planning*, 237, Article 104802. <https://doi.org/10.1016/j.landurbplan.2023.104802>
- Liu, P., Zhao, T., Luo, J., Lei, B., Frei, M., Miller, C., & Biljecki, F. (2023). Towards human-centric digital twins: Leveraging computer vision and graph models to predict outdoor comfort. *Sustainable Cities and Society*, 93, Article 104480. <https://doi.org/10.1016/j.scs.2023.104480>
- Liu, H., Hellín, C., Tayebi, A., Delgado, C., & Gómez, J. (2024). 3D reconstruction of geometries for urban areas supported by computer vision or procedural generations. *Mathematics*, 12(21), 21. <https://doi.org/10.3390/math12213331>
- Luo, J., Zhao, T., Cao, L., & Biljecki, F. (2022a). Semantic riverscapes: Perception and evaluation of linear landscapes from oblique imagery using computer vision. *Landscape and Urban Planning*, 228, Article 104569. <https://doi.org/10.1016/j.landurbplan.2022.104569>
- Luo, J., Zhao, T., Cao, L., & Biljecki, F. (2022b). Water view imagery: Perception and evaluation of urban waterscapes worldwide. *Ecological Indicators*, 145, Article 109615. <https://doi.org/10.1016/j.ecolind.2022.109615>
- Lynda, N., Nnanna, N., & Aminu, A. (2021). Application of deep learning in satellite image-based land cover mapping in Africa. *International Journal of Advanced Computer Science and Applications*, 12(9). <https://doi.org/10.14569/IJACSA.2021.0120948>
- Mai, G., Huang, W., Sun, J., Song, S., Mishra, D., Liu, N., Gao, S., Liu, T., Cong, G., Hu, Y., Cundy, C., Li, Z., Zhu, R., & Lao, N. (2024). On the opportunities and challenges of foundation models for GeoAI. *ACM Transactions on Spatial Algorithms and Systems*, 10(2), 1–46. <https://doi.org/10.1145/3653070>
- Mao, L., Zheng, Z., Meng, X., Zhou, Y., Zhao, P., Yang, Z., & Long, Y. (2022a). Large-scale automatic identification of urban vacant land using semantic segmentation of high-resolution remote sensing images. *Landscape and Urban Planning*, 222, Article 104384. <https://doi.org/10.1016/j.landurbplan.2022.104384>
- Mao, X., Qi, G., Chen, Y., Li, X., Duan, R., Ye, S., He, Y., & Xue, H. (2022b). Towards Robust Vision Transformer. https://openaccess.thecvf.com/content/CVPR2022/html/Mao_Towards_Robust_Vision_Transformer_CVPR_2022_paper.html
- Maquil, V., Leopold, U., De Sousa, L. M., Schwartz, L., & Tobias, E. (2018). Towards a framework for geospatial tangible user interfaces in collaborative urban planning. *Journal of Geographical Systems*, 20(2), 185–206. <https://doi.org/10.1007/s10109-018-0265-6>
- Marasinghe, R., Yigitcanlar, T., Mayere, S., Washington, T., & Limb, M. (2024). Computer vision applications for urban planning: A systematic review of opportunities and constraints. *Sustainable Cities and Society*, 100, Article 105047. <https://doi.org/10.1016/j.scs.2023.105047>
- Matsuzaka, Y., & Yashiro, R. (2023). AI-based computer vision techniques and expert systems. *AI*, 4(1), 1. <https://doi.org/10.3390/ai4010013>
- Metaxiotis, K., Carrillo, J., & Yigitcanlar, T. (2010). *Knowledge-based development for cities and societies: Integrated multi-level approaches*. IGI Global.
- Naik, N., Kominers, S., Raskar, R., Glaeser, E., & Hidalgo, C. (2017). Computer vision uncovers predictors of physical urban change. *Proceedings of the National Academy of Sciences*, 114(29), 7571–7576. <https://doi.org/10.1073/pnas.1619003114>
- Niu, T., Qing, L., Han, L., Long, Y., Hou, J., Li, L., Tang, W., & Teng, Q. (2022). Small public space vitality analysis and evaluation based on human trajectory modeling using video data. *Building and Environment*, 225, Article 109563. <https://doi.org/10.1016/j.buildenv.2022.109563>
- O'Regan, A., Hunter, R., & Nyhan, M. (2021). Biophilic cities: Quantifying the impact of google street view-derived greenspace exposures on socioeconomic factors and self-reported health. *Environmental Science & Technology*, 55(13), 9063–9073. <https://doi.org/10.1021/acs.est.1c01326>
- Payntar, N. D. (2022). Archaeological heritage 'on-the-rise': Detecting emerging tourist attractions in Peru using machine learning and geotagged photographs. *Journal of Heritage Tourism*, 17(2), 222–244. <https://doi.org/10.1080/1743873X.2021.2007254>
- Salazar-Miranda, A., Zhang, F., Sun, M., Leoni, P., Duarte, F., & Ratti, C. (2023). Smart curbs: Measuring street activities in real-time using computer vision. *Landscape and Urban Planning*, 234, Article 104715. <https://doi.org/10.1016/j.landurbplan.2023.104715>
- Sanchez, T. (2023). *Planning with artificial intelligence*. American Planning Association. <https://www.planning.org/publications/report/9270237/>
- Scepanovic, S., Joglekar, S., Law, S., & Quercia, D. (2021). Jane Jacobs in the sky: Predicting urban vitality with open satellite data. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), 48(1–48), 25. <https://doi.org/10.1145/3449257>
- Seneviratne, S., Senanayake, D., Rasnayaka, S., Vidanaarachchi, R., & Thompson, J. (2022). DALLE-URBAN: Capturing the urban design expertise of large text to image transformers. In *2022 International Conference on Digital Image Computing: Techniques and Applications* (pp. 1–9). <https://doi.org/10.1109/DICTAS56598.2022.10034603>
- Sharma, V., Luthra, D., Mann, E., Chaudhary, P., Chowdhary, V., & Jha, C. S. (2022). Change detection and feature extraction using high-resolution remote sensing images. *Remote Sensing in Earth Systems Sciences*, 5(3), 154–164. <https://doi.org/10.1007/s41976-022-00073-6>
- Song, Q., Liu, Y., Qiu, W., Liu, R., & Li, M. (2022a). Investigating the impact of perceived micro-level neighborhood characteristics on housing prices in Shanghai. *Land*, 11(11), 11. <https://doi.org/10.3390/land11112002>
- Song, Y., Ning, H., Ye, X., Chandana, D., & Wang, S. (2022b). Analyze the usage of urban greenways through social media images and computer vision. *Environment and Planning B: Urban Analytics and City Science*, 49(6), 1682–1696. <https://doi.org/10.1177/23998083211064624>
- Stalder, S., Volpi, M., Büttner, N., Law, S., Harttgen, K., & Suel, E. (2024). Self-supervised learning unveils urban change from street-level images. *Computers, Environment and Urban Systems*, 112, Article 102156. <https://doi.org/10.1016/j.compenvurbysys.2024.102156>
- Su, N., Li, W., & Qiu, W. (2023). Measuring the associations between eye-level urban design quality and on-street crime density around New York subway entrances. *Habitat International*, 131, Article 102728. <https://doi.org/10.1016/j.habitatint.2022.102728>
- Suel, E., Muller, E., Bennett, J., Blakely, T., Doyle, Y., Lynch, J., Mackenbach, J., Middel, A., Mizdrak, A., Nathvani, R., Brauer, M., & Ezzati, M. (2023). Do poverty and wealth look the same the world over? A comparative study of 12 cities from five high-income countries using street images. *EPJ Data Science*, 12(1), 1. <https://doi.org/10.1140/epjds/s13688-023-00394-6>
- Sun, L., Feng, L., Zhang, Y., Ji, X., Wong, H., & Wei, Z. (2020). Research on correlation between underground squares' interface morphology and spatial experience based on virtual reality. *International Journal of Pattern Recognition and Artificial Intelligence*, 34(3), Article 2050004. <https://doi.org/10.1142/S0218001420500044>

- Tian, H., Han, Z., Xu, W., Liu, X., Qiu, W., & Li, W. (2021). Evolution of historical urban landscape with computer vision and machine learning: A case study of Berlin. *Journal of Digital Landscape and Architecture*, 16, 436–445.
- Valença, G., Azevedo, C., Moura, F., & Morais de Sá, A. (2025). Creating visualizations using generative AI to guide decision-making in street designs: A viewpoint. *Journal of Urban Mobility*, 7, Article 100104. <https://doi.org/10.1016/j.urbmob.2025.100104>
- Vanky, A., & Le, R. (2023). Urban-semantic computer vision: A framework for contextual understanding of people in urban spaces. *AI & Society*, 38(3), 1193–1207. <https://doi.org/10.1007/s00146-022-01625-6>
- Verma, D., Mumm, O., & Carlow, V. (2021). Identifying streetscape features using VHR imagery and deep learning applications. *Remote Sensing*, 13(17), 17. <https://doi.org/10.3390/rs13173363>
- Wael, S., Elshater, A., & Afifi, S. (2022). Mapping user experiences around transit stops using computer vision technology: Action priorities from Cairo. *Sustainability*, 14(17), 17. <https://doi.org/10.3390/su141711008>
- Wang, Y., Shi, X., Cheng, K., Zhang, J., & Chang, Q. (2022). How do urban park features affect cultural ecosystem services: Quantified evidence for design practices. *Urban Forestry & Urban Greening*, 76, Article 127713. <https://doi.org/10.1016/j.ufug.2022.127713>
- Wang, J., Jiang, L., Yu, H., Feng, Z., Castaño-Rosa, R., & Cao, S. (2024). Computer vision to advance the sensing and control of built environment towards occupant-centric sustainable development: A critical review. *Renewable and Sustainable Energy Reviews*, 192, Article 114165. <https://doi.org/10.1016/j.rser.2023.114165>
- Wang, Q., Liang, Y., Zheng, Y., Xu, K., Zhao, J., & Wang, S. (2025). *Generative AI for urban planning: Synthesizing satellite imagery via diffusion models*. arXiv. <https://doi.org/10.48550/arXiv.2505.08833>.
- Wiley, V., & Lucas, T. (2018). Computer vision and image processing: A paper review. *International Journal of Artificial Intelligence Research*, 2, 22. <https://doi.org/10.29099/ijair.v2i1.42>
- Wong, P., Luo, H., Wang, M., Leung, P., & Cheng, J. (2021). Recognition of pedestrian trajectories and attributes with computer vision and deep learning techniques. *Advanced Engineering Informatics*, 49, Article 101356. <https://doi.org/10.1016/j.aei.2021.101356>
- Wu, P., He, X., Dai, W., Zhou, J., Shang, Y., Fan, Y., & Hu, T. (2025). A review on research and application of AI-based image analysis in the field of computer vision. *IEEE access : Practical innovations, open solutions*, 13, 76684–76702. <https://doi.org/10.1109/ACCESS.2025.3565300>
- Xu, X., Qiu, W., Li, W., Liu, X., Zhang, Z., Li, X., & Luo, D. (2022). Associations between street-view perceptions and housing prices: Subjective vs. objective measures using computer vision and machine learning techniques. *Remote Sensing*, 14(4), 4. <https://doi.org/10.3390/rs14040891>
- Yencha, C. (2019). Valuing walkability: New evidence from computer vision methods. *Transportation Research Part A: Policy and Practice*, 130, 689–709. <https://doi.org/10.1016/j.tra.2019.09.053>
- Yi, S., Liu, X., Li, J., & Chen, L. (2023). UAVformer: A composite transformer network for urban scene segmentation of UAV images. *Pattern Recognition*, 133, Article 109019. <https://doi.org/10.1016/j.patcog.2022.109019>
- Yigitcanlar, T., David, A., Li, W., Fookes, C., Bibri, S., & Ye, X. (2024). Unlocking artificial intelligence adoption in local governments: Best practice lessons from real-world implementations. *Smart Cities*, 7(4), 4. <https://doi.org/10.3390/smartcities7040064>
- Yigitcanlar, T. (2009). Planning for smart urban ecosystems: Information technology applications for capacity building in environmental decision making. *Theoretical and Empirical Researches in Urban Management*, 4(3 (12), 5–21.
- Yuan, J., Zhang, L., & Kim, C. (2023). Multimodal interaction of MU plant landscape design in marine urban based on computer vision technology. *Plants*, 12(7), 7. <https://doi.org/10.3390/plants12071431>
- Yuan, Y., Han, C., Ding, J., Zhang, G., Jin, D., & Li, Y. (2025). *Diffusion transformers as open-world spatiotemporal foundation models*. arXiv. <https://doi.org/10.48550/arXiv.2411.12164>.
- Zhang, C., & Berger, C. (2023). Pedestrian behavior prediction using deep learning methods for urban scenarios: A review. *IEEE Transactions on Intelligent Transportation Systems*, 24(10), 10279–10301. <https://doi.org/10.1109/TITS.2023.3281393>
- Zhang, Y., Ong, G., Jin, Z., Seah, C., & Chua, T. S. (2022). The effects of urban greenway environment on recreational activities in tropical high-density Singapore: A computer vision approach. *Urban Forestry & Urban Greening*, 75, Article 127678. <https://doi.org/10.1016/j.ufug.2022.127678>
- Zhang, L., Han, X., Wu, J., & Wang, L. (2023). Mechanisms influencing the factors of urban built environments and coronavirus disease 2019 at macroscopic and microscopic scales: The role of cities. *Frontiers in Public Health*, 11, 1–10. <https://doi.org/10.3389/fpubh.2023.1137489>
- Zhang, F., Salazar-Miranda, A., Duarte, F., Vale, L., Hack, G., Chen, M., Liu, Y., Batty, M., & Ratti, C. (2024). Urban visual intelligence: Studying cities with artificial intelligence and street-level imagery. *Annals of the American Association of Geographers*, 114(5), 876–897. <https://doi.org/10.1080/24694452.2024.2313515>
- Zhang, W., Han, J., Xu, Z., Ni, H., Lyu, T., Liu, H., & Xiong, H. (2025). *Towards urban general intelligence: A review and outlook of urban foundation models*. arXiv. <https://doi.org/10.48550/arXiv.2402.01749>.
- Zhao, C., Ogawa, Y., Chen, S., Oki, T., & Sekimoto, Y. (2023). Quantitative land price analysis via computer vision from street view images. *Engineering Applications of Artificial Intelligence*, 123, Article 106294. <https://doi.org/10.1016/j.engappai.2023.106294>
- Zhao, Y. (2024). The creative cosmos beyond humans: A symphony of participatory design and visual artificial intelligence. *Frontiers in Communication*, 9, 1–10. <https://doi.org/10.3389/fcomm.2024.1440384>
- Zhou, T., Xia, W., Zhang, F., Chang, B., Wang, W., Yuan, Y., Konukoglu, E., & Cremers, D. (2024). *Image segmentation in foundation model era: A survey*. arXiv. <https://doi.org/10.48550/arXiv.2408.12957>.
- Zou, J., Kim, B., Kim, H., & Al-Husseini, M. (2012). An automated system for the creation of an urban infrastructure 3D model using image processing techniques. *KSCE Journal of Civil Engineering*, 16(1), 9–17. <https://doi.org/10.1007/s12205-012-1272-7>